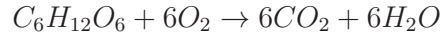


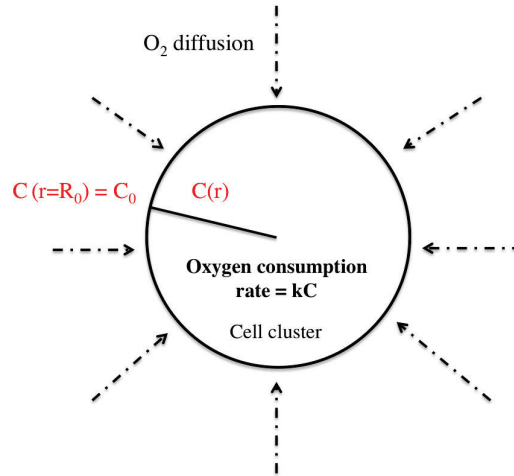
Question 1: Cellular respiration in cells

The cells in human body require a source of energy to function and they are able to obtain it by breaking down food molecules to release the stored energy inside. This process is called “cellular respiration”. Oxygen is consumed to break down food (e.g. glucose) in this process.



Assume a constant concentration of glucose inside the cells and its concentration is much more than the oxygen concentration. Therefore, the reaction rate can be approximated to be a 1st order reaction with respect to oxygen concentration. In this question, we will be looking at the transport and consumption of oxygen inside a cluster of cells.

We consider a steady-state O_2 diffusion in a sphere of radius R_0 , with a irreversible reaction of O_2 consumption. Assume that the oxygen concentration at the outer surface of the cell cluster is maintained at C_0 where the value of rate constant k is $10^{-2}/s$, the spherical radius is 0.02 m , the diffusivity D is $10^{-5}\text{ m}^2/s$ and C_0 is taken to be $5\text{ mol}/\text{m}^3$.



(a) By using shell balance technique, derive the governing equation and the boundary conditions for oxygen concentration profile, $C(r)$. State any assumptions you have made.

- (b) Scale the variables to non-dimensionalize the governing equation and boundary conditions. Discuss the physical meaning of the generated dimensionless parameters.
- (c) Solve the above ODE to determine $C(r)$ and use Matlab to plot $C(r)$.
- (d) If R_0 and reaction rate constant k are sufficiently large, not enough oxygen will reach a central core, defined by $r < R_{core}$. This situation occurs in certain tumors, where as the tumor grows, the cells in the centre are killed due to lack of oxygen. The critical concentration, below which the cell will be killed is 0.25 mol/m^3 . Determine whether this situation will occur. Briefly discuss what are the impact of R_0 and k on R_{core} based on the results/plots obtained.
- (e) Can we simplify the model and get away from Bessel function? Explain. (Hint: if we assume zeroth-order oxygen consumption rate and simplify the model, would our solution be reasonably accurate?)

Learning Outcomes:

1. How to model a process and the use of dimensionless parameter to derive simple governing equation.
2. Interpretation of the dimensionless parameters' physical meaning.
3. Concepts of Bessel equation and functions. How to solve Bessel equation.
4. Discussion of system parameters' impact on system behaviour
5. Discussion on validity and limitations on assumptions made. Learn the impact of such assumptions.
6. Use of MATLAB to plot. Find roots of complex functions through a graphical method with the aid of MATLAB.

Solution to Question 1:

(a) The general shell balance equation in any control volume has the form

$$Accumulation = In - Out + Generation - Consumption$$

and in this question

$$In : (-D \frac{dC}{dr} 4\pi r^2)|_r \quad (1)$$

$$Out : (-D \frac{dC}{dr} 4\pi r^2)|_{r+\Delta r} \quad (2)$$

$$Generation : 0 \quad (3)$$

$$Consumption : kC\Delta V = kC4\pi r^2\Delta r \quad (4)$$

$$Accumulation : 0 \text{ (assume steady-state)} \quad (5)$$

Substitute the above 5 equations into shell balance equation and we obtain

$$\frac{(D \frac{dC}{dr} 4\pi r^2)|_{r+\Delta r} - (D \frac{dC}{dr} 4\pi r^2)|_r}{\Delta r} = kC4\pi r^2 \quad (6)$$

By taking the limit as $\Delta r \rightarrow 0$,

$$D \frac{d}{dr}(r^2 \frac{dC}{dr}) - kCr^2 = 0 \quad (7)$$

The boundary conditions are

$$\frac{dC}{dr}(0) = 0 \quad (8)$$

$$C(R_0) = 0 \quad (9)$$

(b) Nondimensionalization: let $\theta = \frac{C}{C_0}$ and $\eta = \frac{r}{R_0}$. The equation (7) can be transformed into

$$\frac{1}{\eta^2} \frac{d}{d\eta}(\eta^2 \frac{d\theta}{d\eta}) - \frac{kR_0^2}{D} \theta = 0 \quad (10)$$

Let $Da = \frac{kR_0^2}{D}$ and the above equation becomes

$$\frac{1}{\eta^2} \frac{d}{d\eta}(\eta^2 \frac{d\theta}{d\eta}) - Da \theta = 0 \quad (11)$$

with corresponding transformed boundary conditions

$$\frac{d\theta}{d\eta}(0) = 0 \quad (12)$$

$$\theta(1) = 1 \quad (13)$$

Here Da is a dimensionless number called Damkhler number, which is used to relate the chemical reaction timescale (reaction rate) to the transport phenomena rate occurring in a system.

$$Da = \frac{kR_0^2}{D} = \frac{kC_0(4\pi R_0^3)}{D(\frac{C_0}{R_0})4\pi R_0^2} \sim \frac{\text{Reaction consumption rate}}{\text{Diffusion rate}} \quad (14)$$

(c) The expansion of equation (11) gives

$$\eta^2 \frac{d^2\theta}{d\eta^2} + 2\eta \frac{d\theta}{d\eta} - Da \eta^2 \theta = 0 \quad (15)$$

This is a modified spherical Bessel's equation and the general solution is

$$\theta(\eta) = C_1 \frac{\sinh(\sqrt{Da}\eta)}{\eta} + C_2 \frac{\cosh(\sqrt{Da}\eta)}{\eta} \quad (16)$$

To keep the concentration finite at $\eta = 0$ and also to satisfy the no-flux condition, it is necessary that $C_2 = 0$. Alternatively, we can show it by using the Taylor series expansion of $\sinh$ and $\cosh$,

$$\cosh(\sqrt{Da}\eta) = 1 + \frac{(\sqrt{Da}\eta)^2}{2} + O((\sqrt{Da}\eta)^4) \quad (17)$$

$$\sinh(\sqrt{Da}\eta) = x + \frac{(\sqrt{Da}\eta)^3}{6} + O((\sqrt{Da}\eta)^5) \quad (18)$$

When we differentiate equation (16) with respect to η and then replace $\sinh$ and $\cosh$ with equation (17) and (18), we obtain

$$\frac{d\theta}{d\eta} = C_1 \frac{1}{3} Da^{\frac{3}{2}} \eta + C_2 \left(-\frac{3}{2} Da - \frac{1}{\eta^2} - \frac{1}{6} Da^2 \eta^2 \right) + O(\eta^4) \quad (19)$$

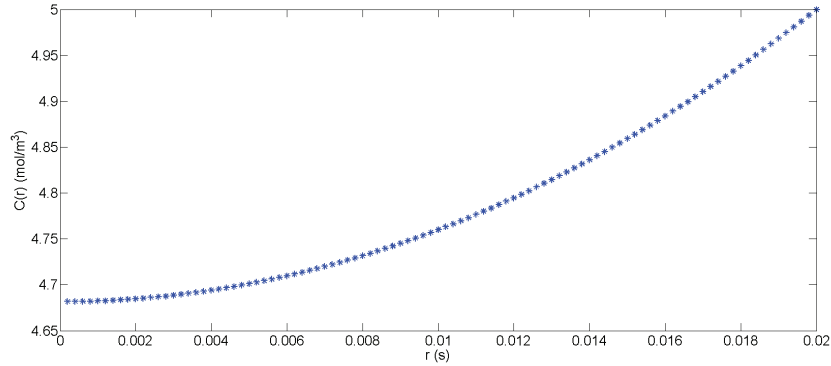
When $\eta \rightarrow 0$, $\frac{d\theta}{d\eta} \rightarrow 0$. This forces $C_2 = 0$, otherwise $\frac{d\theta}{d\eta}$ will blow up. Therefore, by evaluating C_1 using the boundary condition at the surface, the solution is

$$\theta(\eta) = \frac{\sinh(\sqrt{Da}\eta)}{\eta \sinh(\sqrt{Da})} \quad (20)$$

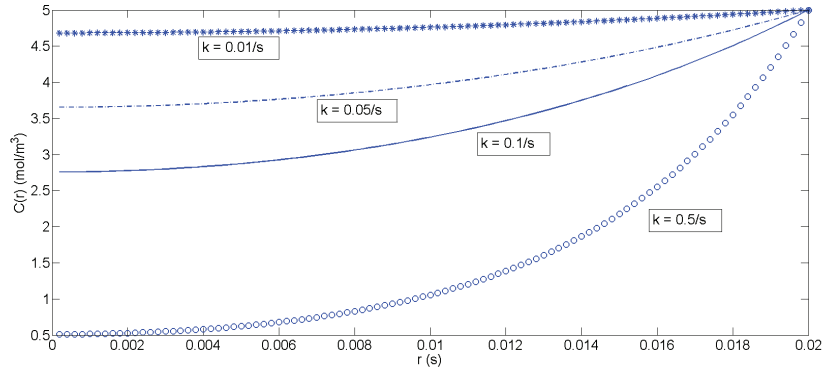
and

$$C(r) = \frac{C_0 \sinh(\sqrt{Da} \frac{r}{R_0})}{\frac{r}{R_0} \sinh(\sqrt{Da})} \quad (21)$$

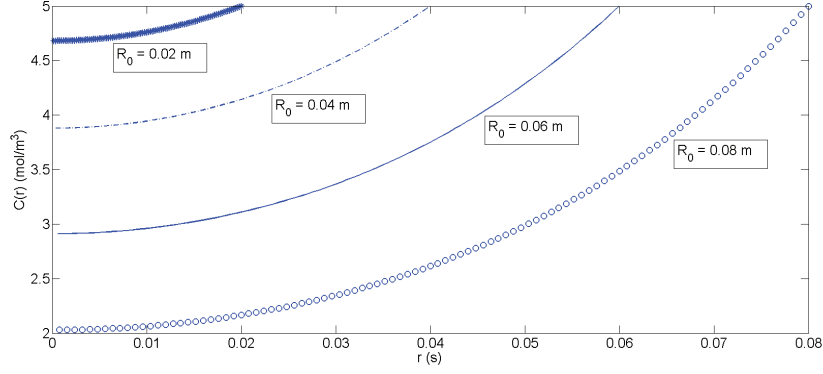
Now, we can use Matlab to plot $C(r)$, as shown in the graph below. Moving more and more inside of the cluster of cells, the oxygen concentration is decreasing.



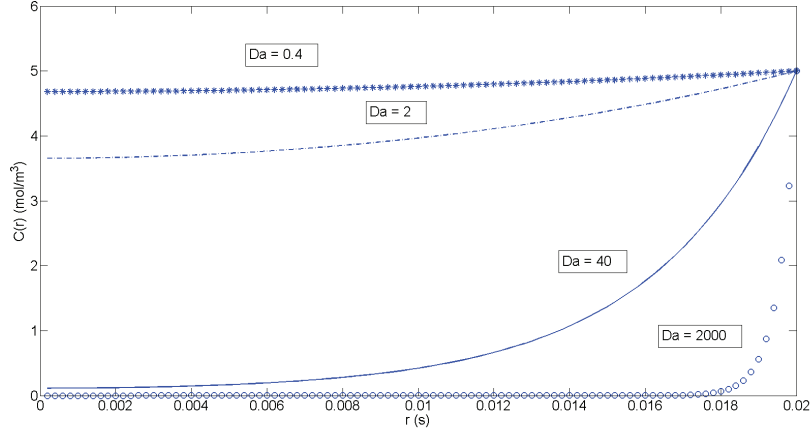
Let us take a closer look at the effect of k , R_0 and Da on the oxygen concentration profile now. Firstly, let us look at the reaction rate constant k . As indicated in the figure below, with increasing k , thus faster oxygen consumption rate inside the cell cluster, the oxygen concentration drops faster.



Next, let us look at the effect of R_0 . As shown in the figure below, there are two things worth noting. (i) As the cluster size increases, the consumption of oxygen becomes faster, and (ii) as the cluster size increases, oxygen concentration at the core of the cluster decreases, and with a large enough radius, it might drop to 0.

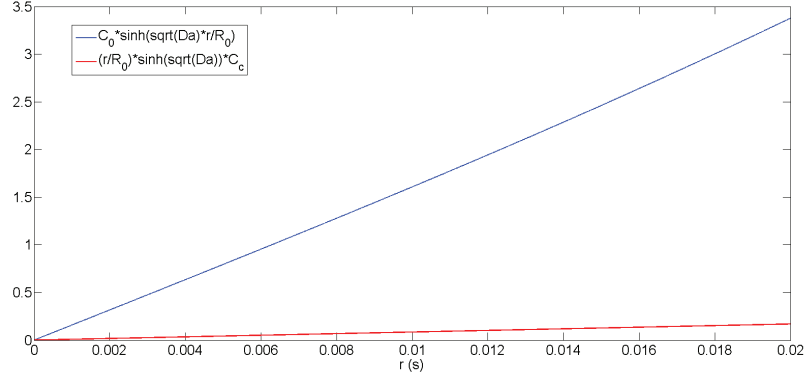


Lastly, let us look at Da . As discussed in part (b), Da is a measure of the ratio between reaction consumption rate and diffusion rate. The larger Da is, the more dominant the reaction consumption is, thus the faster the depletion of oxygen inside the cluster. This is consistent with the plots shown below.

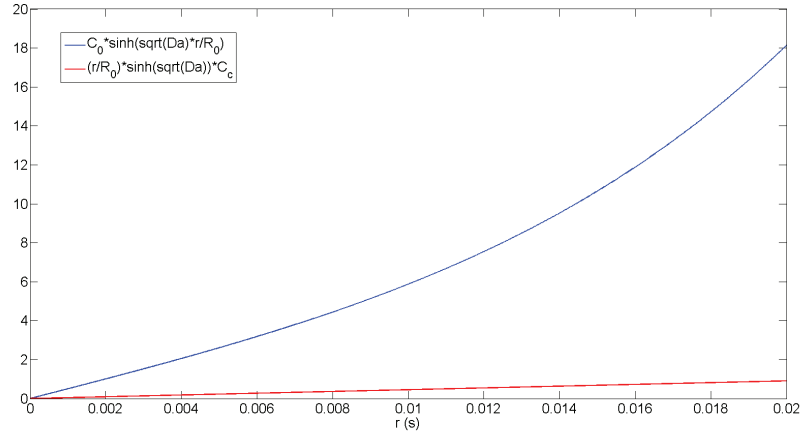


(d) As obtained in part (c), $C(r) = \frac{C_0 \sinh(\sqrt{Da} \frac{r}{R_0})}{\frac{r}{R_0} \sinh(\sqrt{Da})}$. As we already saw in the plots in part (c), as moving inside the cell cluster, concentration of oxygen decreases. Thus, to find R_{core} , we just need to find the intersection of two curves: (Matlab codes are attached at the end) (i) $C_0 \sinh(\sqrt{Da} \frac{r}{R_0})$ and (ii) $\frac{r}{R_0} \sinh(\sqrt{Da}) \cdot C_c$ where C_c is the critical concentration. First let us look at the case described in the question. The plot below indicates that there is no

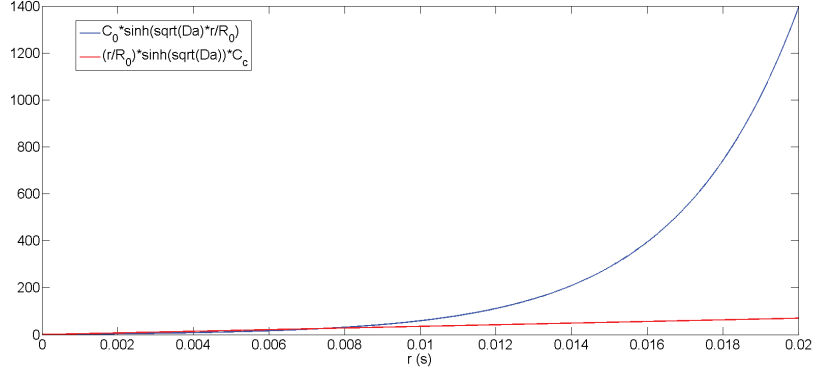
intersection of the two curves thus the oxygen depletion situation will not occur.



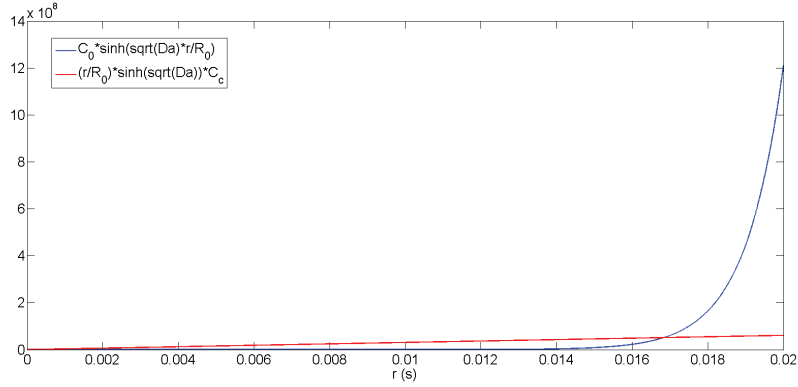
Now, let us see what happens when the reaction consumption rate is increased. As shown in the figures below, the oxygen depletion will start to occur as k increases, and R_{core} moves closer and closer to the surface of the cluster. Note that here R_0 is maintained at 0.02 m .



$k = 0.1/s$, no intersection

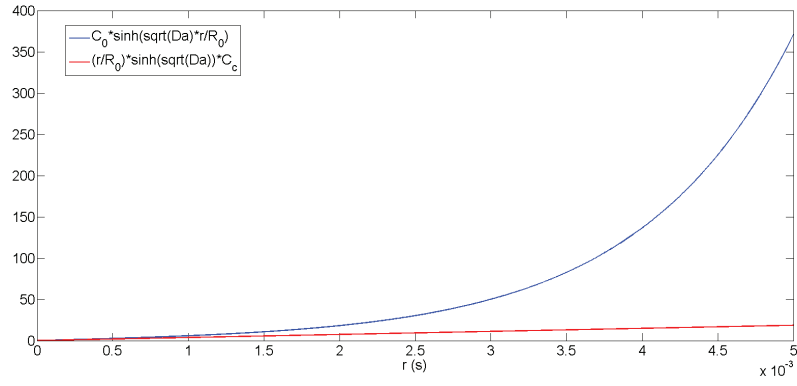


$$k = 1/s, R_{core} = 0.0074 m$$

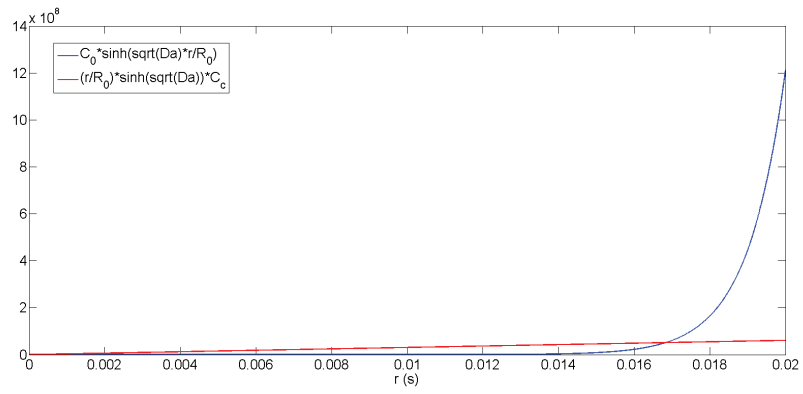


$$k = 10/s, R_{core} = 0.0168 m$$

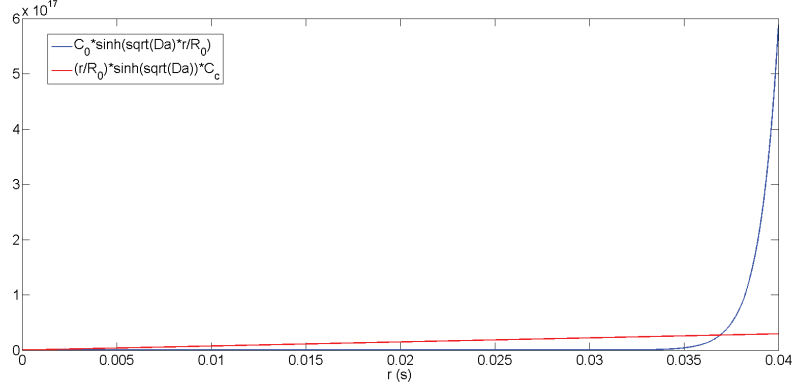
Lastly, if we keep k as a constant ($= 10/s$) and vary R_0 , as indicated in the figures below, R_0 seems to have similar impact as k . With increasing R_0 , R_{core} starts to appear and moves towards the surface of the cluster. Note that it seems that the increase of R_0 has a higher impact than k , probably due to their different importance level in Da (Recall that $Da = \frac{kR_0^2}{D}$).



$R_0 = 0.005 \text{ m}$, no intersection



$R_0 = 0.02 \text{ m}$, $R_{core} = 0.0168 \text{ m}$



$$R_0 = 0.04 \text{ m}, R_{core} = 0.0369 \text{ m}$$

(e) If we assume a constant reaction rate of oxygen, which is independent on the concentration of oxygen, the shell balance equation then becomes

$$D \frac{d}{dr} \left(r^2 \frac{dC}{dr} \right) - K_c r^2 = 0 \quad (22)$$

where K_c represents the constant consumption rate of oxygen. The above equation can be solved by direct integration after moving $K_c r^2$ to the left hand side and the solution is in the form

$$C(r) = \frac{K_c}{6D} r^2 - \frac{C_3}{rD} + C_4 \quad (23)$$

With the boundary conditions from equation (8) and (9), we can obtain

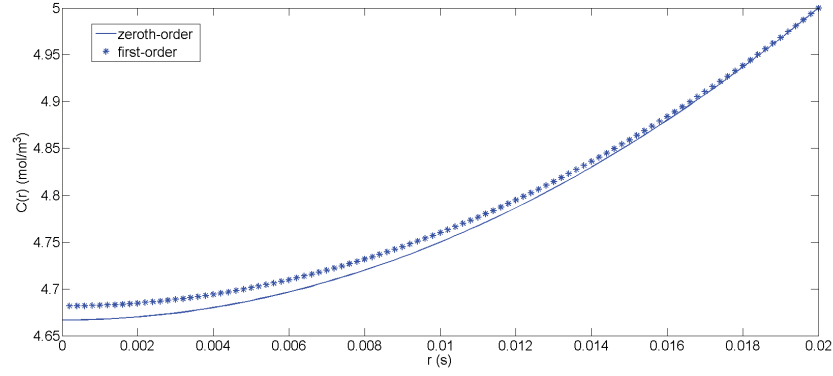
$$C(r) = \frac{K_c}{6D} r^2 + C_0 - \frac{K_c R_0^2}{6D} \quad (24)$$

Now if we assume $K_c = kC_0$ as a guidance and calculate for K_c

$$K_c = \frac{0.01}{s} \cdot 5 \frac{\text{mol}}{m^3} = 0.05 \frac{\text{mol}}{s m^3}$$

We can then use Matlab to plot and compare the profile with the 1st-order reaction case. (See code attached at the end of the answer)

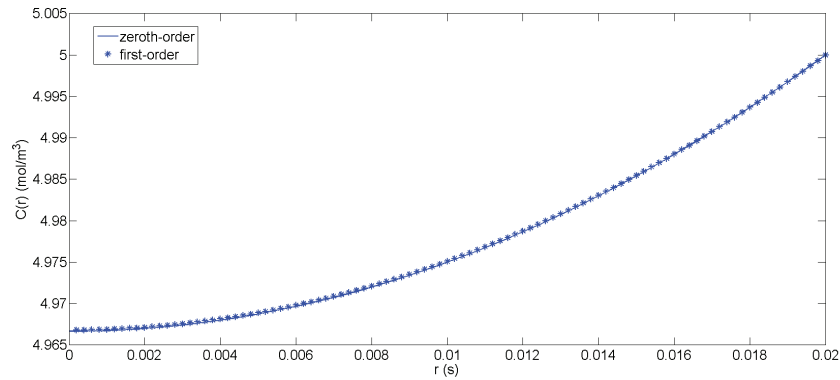
As shown in the figure below, the deviation is not obvious at positions closer to the surface of the cluster, and grows when moving more and more inside the cell cluster.



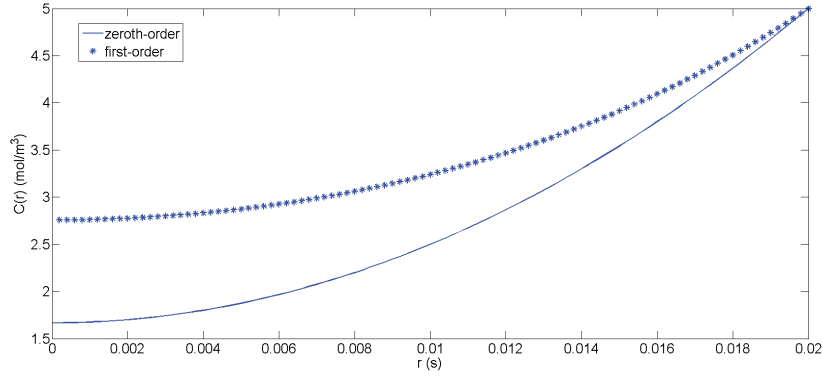
Let us then look at what are the impacts of k and R_0 on the deviation (note that when k is varied, R_0 is kept constant and vice versa).

The impact of k

As shown in the figured below, when k is very small, the concentration profile obtained from both zeroth-order and first-order almost exactly matches each other, indicating that our assumption is reasonably valid when k is very small. While when k becomes much larger, the deviation grows as well, and the two models deviate by around 40% for the core concentration when k is 100 times larger.



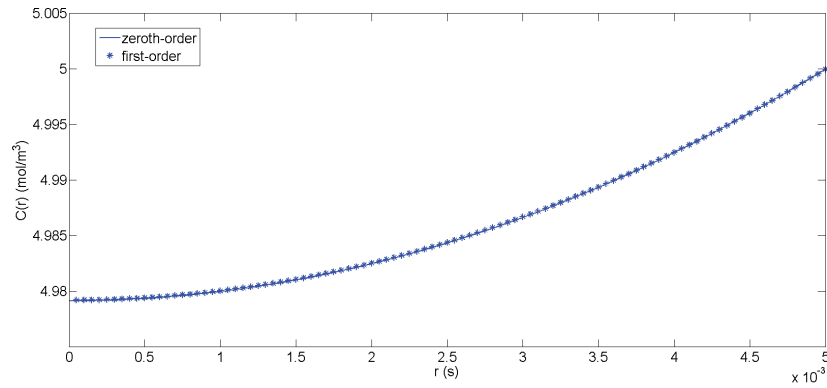
$$k = 0.001$$



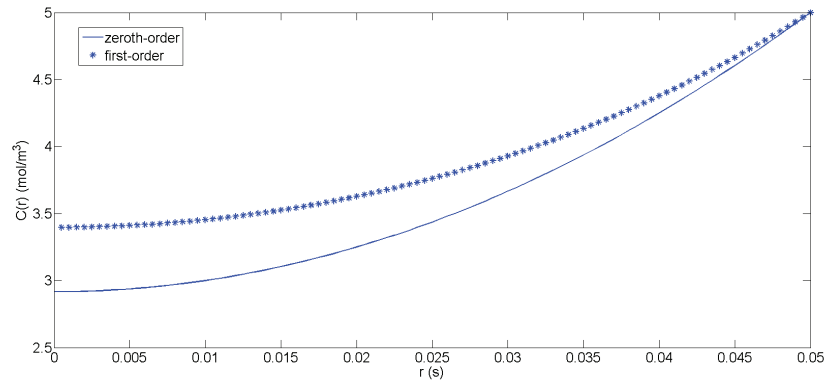
$$k = 0.1$$

The impact of R_0

As shown in the figure below, when R_0 is very small, the concentration profile obtained from both zeroth-order and first-order almost exactly matches each other, indicating that our assumption is reasonably valid when the size of the cell cluster is very small. While when R_0 becomes much larger, the deviation grows as well.



$$R_0 = 0.005 m$$



$$R_0 = 0.05 \text{ m}$$

Matlab codes for Question 1

For (c)

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
k=10^(-2); %1/s
R_0=2*10^(-2);%m
C_0=5;%mol/m^3
D=10^(-5); %m^2/s
Da=k*R_0^2/D;
r=0:(R_0/100):R_0;

C_original=C_0*sinh(sqrt(Da)*r/R_0)./(r/R_0)/sinh(sqrt(Da));
plot(r,C_original,'*');
xlabel('r (s)')
ylabel('C(r) (mol/m^3)')
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% for Q2c_plot_compare_Da
k=10^(-2); %1/s
R_0=2*10^(-2);%m
C_0=5;%mol/m^3
D=10^(-5); %m^2/s
Da=k*R_0^2/D;
r=0:(R_0/100):R_0;

C_original=C_0*sinh(sqrt(Da)*r/R_0)./(r/R_0)/sinh(sqrt(Da));
```

```

plot(r,C_original,'*');
xlabel('r (s)')
ylabel('C(r) (mol/m^3)')

hold on

k=5*10^(-2); %1/s
R_0=2*10^(-2);%m
D=10^(-5);
Da=k*R_0^2/D;

C_1=C_0*sinh(sqrt(Da)*r/R_0)./(r/R_0)/sinh(sqrt(Da));
plot(r,C_1,'-');

k=10^(-1); %1/s
R_0=2*10^(-2);%m
D=10^(-6);
Da=k*R_0^2/D;

C_2=C_0*sinh(sqrt(Da)*r/R_0)./(r/R_0)/sinh(sqrt(Da));
plot(r,C_2,'-');

k=5*10^(-1); %1/s
R_0=2*10^(-2);%m
D=10^(-7);
Da=k*R_0^2/D;

C_3=C_0*sinh(sqrt(Da)*r/R_0)./(r/R_0)/sinh(sqrt(Da));
plot(r,C_3,'o');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% for Q2c_plot_compare_k
k=10^(-2); %1/s
R_0=2*10^(-2);%m
C_0=5;%mol/m^3
D=10^(-5); %m^2/s
Da=k*R_0^2/D;
r=0:(R_0/100):R_0;

```

```

C_original=C_0*sinh ( sqrt (Da)*r/R_0 )./( r/R_0)/ sinh ( sqrt (Da));
plot (r , C_original , ' * ' );
xlabel ( ' r   ( s ) ' )
ylabel ( ' C(r)   (mol/m^3) ' )

hold on

k=5*10^(-2); %1/s
R_0=2*10^(-2);%m

Da=k*R_0^2/D;

C_1=C_0*sinh ( sqrt (Da)*r/R_0 )./( r/R_0)/ sinh ( sqrt (Da));
plot (r , C_1 , ' - . ' );

k=10^(-1); %1/s
R_0=2*10^(-2);%m

Da=k*R_0^2/D;

C_2=C_0*sinh ( sqrt (Da)*r/R_0 )./( r/R_0)/ sinh ( sqrt (Da));
plot (r , C_2 , ' - ' );

k=5*10^(-1); %1/s
R_0=2*10^(-2);%m

Da=k*R_0^2/D;

C_3=C_0*sinh ( sqrt (Da)*r/R_0 )./( r/R_0)/ sinh ( sqrt (Da));
plot (r , C_3 , ' o ' );
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% for Q2c_plot_compare_R0
k=10^(-2); %1/s
R_0=2*10^(-2);%m
C_0=5;%mol/m^3
Da=k*R_0^2/D;
D=10^(-5); %m^2/s
r=0:(R_0/100):R_0;

```

```

C_original=C_0*sinh(sqrt(Da)*r/R_0)./(r/R_0)/sinh(sqrt(Da));
plot(r,C_original,'*');
xlabel('r (s)')
ylabel('C(r) (mol/m^3)')

hold on

k=10^(-2); %1/s
R_0=4*10^(-2);%m
r=0:(R_0/100):R_0;
Da=k*R_0^2/D;

C_1=C_0*sinh(sqrt(Da)*r/R_0)./(r/R_0)/sinh(sqrt(Da));
plot(r,C_1,'-');

k=10^(-2); %1/s
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
R_0=6*10^(-2);%m
r=0:(R_0/100):R_0;
Da=k*R_0^2/D;

C_2=C_0*sinh(sqrt(Da)*r/R_0)./(r/R_0)/sinh(sqrt(Da));
plot(r,C_2,'-');

k=10^(-2); %1/s
R_0=8*10^(-2);%m
r=0:(R_0/100):R_0;
Da=k*R_0^2/D;

C_3=C_0*sinh(sqrt(Da)*r/R_0)./(r/R_0)/sinh(sqrt(Da));
plot(r,C_3,'o');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

For (d)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
clear all;clc;
k=0.01; %1/s this can be varied to make the comparison
R_0=2*10^(-2);%m this can be varied to make the comparison
C_0=5;%mol/m^3

```



```

C_c=0.25; %critical concentration
D=10(-5); %m2/s
r=0:(R_0/1000):R_0;

```

```

Da=k*R_02/D;

```

```

figure(2);
y1=C_0*sinh(sqrt(Da)*r/R_0);
y2=(r/R_0)*sinh(sqrt(Da))*C_c;
plot(r,y1,'b',r,y2,'r');
xlabel('r (s)')

```

```

[xi,yi] = polyxpoly(r,y1,r,y2);
%to find the intersection of the two curves
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

```

For (e)

```

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% for plot the 0th order and 1st order solution
clear all
k=10(-2); %1/s this can be varied for other plots in Q2e
R_0=2*10(-2); %m this can be varied for other plots in Q2e
C_0=5; %mol/m3
K_c=k*C_0;
D=10(-5); %m2/s
r=0:(R_0/100):R_0;
%0th order concentration profile
C=(K_c/6/D)*r.2+C_0-K_c*R_02/6/D;
plot(r,C)
hold on
xlabel('r (s)')
ylabel('C(r) (mol/m3)')
%add in the part for Q2c plot%
Da=k*R_02/D;
C_original=C_0*sinh(sqrt(Da)*r/R_0)./(r/R_0)/sinh(sqrt(Da));
%1st order concentration profile
plot(r,C_original,'*');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

```